Srithvik Ellanki

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Authorized to Handle CUI

EDUCATION

Florida Institute of Technology **Aug 2024 - May 2027**
B.S. Aerospace Engineering (Astronautics Emphasis) | 3.38 GPA | Dean's List | Maximum Panther Fund Scholarship

EXPERIENCE

Computer Vision Engineer | U.S. Navy (CUI) **Apr 2026 - Present**
- Built a Python computer vision localization system that ingests detected object data and outputs a real-time positional graph of estimated heading, bearing, and geographic location, each reported within a defined confidence tolerance
- Reached localization accuracy within 0.1 km of a known position at 1–1.5 seconds per processing cycle, validated through one real-world test and two staged test setups, with iterative debugging across edge cases in the detection data
- Independently architected the full pipeline from an inherited codebase, designed ahead of live hardware integration
- Delivered 3 quarterly progress review presentations and multiple technical reports to Navy personnel and USC and partner university stakeholders, covering design decisions, testing methodology, and performance benchmarks

Project Manager | CubeSat Senior Design Team **Jan 2026 - Present**
- Led a 10-person team across four subsystems (Attitude Determination and Control, Structures, Electrical Power, Payload), coordinating project milestones and system-level performance goals for a flight-ready 6U CubeSat
- Managed project timelines, task allocation, and design reviews using ABET-style planning and risk mitigation
- Contributed to Structures through CAD, integration, and FEA validation of the stacked solar panel deployment
- Directed system verification, subsystem interface checks, and data analysis to ensure mission readiness

Lead Propulsion Engineer | AIAA Panther Rocket Labs **Aug 2024 - Feb 2026**
- Trained and led a large multidisciplinary team in CAD, FEA, CFD, CNC machining, and solid propellant handling
- Designed, simulated, and integrated a complete solid propulsion system targeting a 100 km apogee, conducting thermal, structural, and nozzle erosion analyses using FEA alongside internal ballistics modeling of the motor
- Directed 7 consecutive successful static fires, establishing the university's first verified propulsion test campaign while improving safety protocols and raising validation standards through thorough documentation of every test
- Ran hydrostatic proof tests on motor cases and added an assembly seating check after an early hydrostatic anomaly

Undergraduate Researcher | FIT - School of Psychology **Jan 2025 - Aug 2025**
- Studied Tetris and Pac-Man decision-making with Baylor Medicine across ~100 patients for epilepsy research
- Wrote MATLAB scripts to automate video processing and improve efficiency, cutting overall analysis time by 50%
- Presented findings at the Northrop Grumman Showcase, highlighting applications in predictive analytics
- Processed and analyzed patient task recordings in MATLAB to quantify decision-making patterns across study groups

Aerospace Engineering Lecturer | Envision by WorldStrides, Rice University **May 2026 - Aug 2026**
- Taught propulsion, stability, avionics and near space flight to 40 high school students through hands-on builds
- Led small scale rocket launches and coached student teams through drone flight computer wiring and debugging
- Debriefed each team after launch, explaining how thrust, stability, and fin placement shaped their flight results
- Supported high altitude weather balloon launches with scientific payloads in partnership with NASA Education

PROJECTS

Phantom 10k - Solid Rocket Motor | *OpenMotor, PROPEP, Inventor, MATLAB, CNC* **Oct 2025**
- Led team design and development of a flight-ready solid rocket motor, coordinating propellant formulation, grain geometry optimization, nozzle design, and system integration across the full build cycle, from initial concept to flight
- Applied internal ballistics, nozzle theory, and grain regression analysis in OpenMotor and PROPEP to iteratively predict, simulate, and experimentally verify motor performance across 3 motor configurations ahead of the flight build
- Built and flew a K521 motor (348.12 lbf·s) with clean ignition, a stable 500 psi pressure curve, and reliable data
- Wrote the failure analysis for a static fire case rupture, tracing it to backside burning and revising the grain bonding

Project Aurelian - Bipropellant Engine| *Fusion, Ansys, CNC* **May 2026 - Aug 2026**
- Designed a bipropellant engine in Fusion, sizing the injector, chamber, and nozzle for 360 lbf on nitrous and IPA
- Ran Ansys structural and thermal analyses confirming the chamber and nozzle withstand 500 psi and combustion heat
- Prepared the design for CNC machining, translating CAD models into producible engine components
- Served as safety coordinator, verifying designs met specifications and enforcing engineering protocols

SKILLS

Programming & Tools: Python (OpenCV, NumPy), MATLAB, C++, Git, MS Office
CAD / Simulation: Siemens NX, AutoCAD, Inventor, Fusion, Ansys, OpenRocket, OpenMotor, PROPEP
Analysis: FEA (structural/thermal), CFD, Internal Ballistics, Nozzle Performance
Fabrication: CNC Machining, Solid Propellant Handling, Structural Testing
Coursework: Space Flight Mechanics, Space Vehicle Control, Thermodynamics, Rockets and Mission Analysis